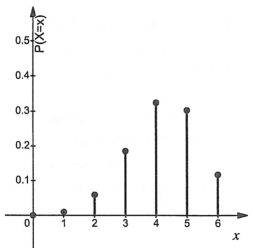
The Binomial Distribution

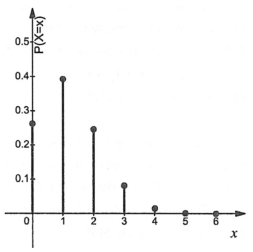
Questions 1–63

Question 1. The four diagrams below show $X \sim \mathrm{B}(6, 0.2)$, $X \sim \mathrm{B}(6, 0.5)$, $X \sim \mathrm{B}(6, 0.7)$ and $X \sim \mathrm{B}(6, 0.89)$. Which is which?

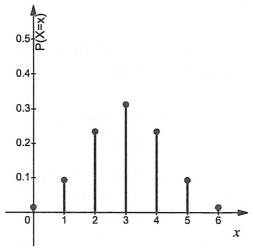
A



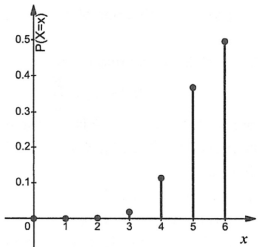
B



C



D



Question 2. Consider the stick graph of the following distributions. Do they show a positive skew, a negative skew, or are they symmetric?

a) $X \sim \mathrm{B}(45, 0.3)$
b) $X \sim \mathrm{B}(32, 0.5)$
c) $X \sim \mathrm{B}(24, 0.9)$
d) $X \sim \mathrm{B}(60, 0.51)$

Question 3. Given $X \sim \mathrm{B}(3, 0.2)$, copy and complete the table shown below and draw a stick graph to represent the distribution.

x	0	1	2	3
$\mathrm{P}(X = x)$				

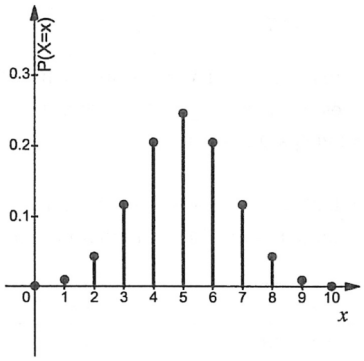
Question 4. Given $X \sim \mathrm{B}(3, 0.6)$, copy and complete the table shown below and draw a stick graph to represent the distribution.

x	0	1	2	3
$\mathrm{P}(X = x)$				

Question 5. Given $X \sim \mathrm{B}(4, 0.5)$, copy and complete the table shown below and draw a stick graph to represent the distribution.

x	0	1	2	3	4
$\mathrm{P}(X = x)$					

Question 6. The diagram below shows $X \sim \mathrm{B}(n, p)$. Write down n and suggest a value for p



Question 7. Given $X \sim \mathrm{B}(15, 0.3)$, **use the formula** to find:

a) $\mathrm{P}(X = 3)$
b) $\mathrm{P}(X = 4)$
c) $\mathrm{P}(X = 5)$
d) $\mathrm{P}(X = 9)$

Question 8. Use the formula in each case.

a) Given $X \sim \mathrm{B}(10, 0.6)$, find $\mathrm{P}(X = 6)$
b) Given $X \sim \mathrm{B}(25, 0.2)$, find $\mathrm{P}(X = 4)$
c) Given $X \sim \mathrm{B}(60, 0.9)$, find $\mathrm{P}(X = 57)$
d) Given $X \sim \mathrm{B}(105, 0.56)$, find $\mathrm{P}(X = 52)$

Question 9. Given $X \sim \mathrm{B}(10, 0.7)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X = 5)$

b) $\mathrm{P}(X = 7)$

c) $\mathrm{P}(X = 10)$

Question 10. Given $X \sim \mathrm{B}(20, 0.2)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X = 1)$

b) $\mathrm{P}(X = 3)$

c) $\mathrm{P}(X = 5)$

Question 11. Given $X \sim \mathrm{B}(17, 0.16)$, **use the formula** to find:

a) $\mathrm{P}(X \leq 1)$

b) $\mathrm{P}(X \leq 2)$

c) $\mathrm{P}(X \leq 3)$

Question 12. Given $X \sim \mathrm{B}(22, 0.35)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X \leq 2)$

b) $\mathrm{P}(X \leq 5)$

c) $\mathrm{P}(X \leq 8)$

Question 13. Use your calculator's binomial probability function in each case.

a) Given $X \sim \mathrm{B}(18, 0.24)$, find $\mathrm{P}(X \leq 5)$
b) Given $X \sim \mathrm{B}(27, 0.36)$, find $\mathrm{P}(X \leq 15)$
c) Given $X \sim \mathrm{B}(39, 0.49)$, find $\mathrm{P}(X \leq 20)$
d) Given $X \sim \mathrm{B}(47, 0.61)$, find $\mathrm{P}(X \leq 25)$

Question 14. Given $X \sim \mathrm{B}(18, 0.68)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X < 9)$
b) $\mathrm{P}(X < 11)$
c) $\mathrm{P}(X < 14)$

Question 15. Given $X \sim \mathrm{B}(17, 0.49)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X \geq 8)$

b) $\mathrm{P}(X \geq 11)$

c) $\mathrm{P}(X \geq 12)$

Question 16. Use your calculator's binomial probability function in each case.

a) Given $X \sim \mathrm{B}(16, 0.37)$, find $\mathrm{P}(X \geq 4)$

b) Given $X \sim \mathrm{B}(38, 0.02)$, find $\mathrm{P}(X \geq 2)$

c) Given $X \sim \mathrm{B}(71, 0.77)$, find $\mathrm{P}(X \geq 52)$

Question 17. Given $X \sim \mathrm{B}(23, 0.66)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(9 \leq X \leq 12)$

b) $\mathrm{P}(10 \leq X \leq 14)$

c) $\mathrm{P}(12 \leq X \leq 14)$

Question 18. Given $X \sim \mathrm{B}(18, 0.24)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(2 < X \leq 3)$

b) $\mathrm{P}(3 < X \leq 7)$

c) $\mathrm{P}(4 < X \leq 10)$

Question 19. Given $X \sim \mathrm{B}(29, 0.49)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(14 \leq X < 17)$

b) $\mathrm{P}(13 \leq X < 18)$

c) $\mathrm{P}(10 \leq X < 13)$

Question 20. Given $X \sim \mathrm{B}(13, 0.45)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(4 < X < 6)$

b) $\mathrm{P}(5 < X < 10)$

c) $\mathrm{P}(6 < X < 9)$

Question 21. In a large population of a certain type of tropical bird, 20% are known to have yellow feathers. A random sample of 30 of these birds are selected. Calculate the probability that the sample contains exactly 5 with yellow feathers.

Question 22. A factory produces blades for pencil sharpeners. The probability of one of these blades being misshapen is 0.008. In a random sample of 200 blades, find the probability of less than 3 being misshapen.

Question 23. Write down the mean, $\mathrm{E}(X)$, and variance, $\mathrm{Var}(X)$ of the following distributions.

a) $X \sim \mathrm{B}(8, 0.5)$
b) $X \sim \mathrm{B}(20, 0.3)$
c) $X \sim \mathrm{B}(16, 0.9)$
d) $X \sim \mathrm{B}(25, 0.2)$
e) $X \sim \mathrm{B}(40, 0.6)$

Question 24. Given $X \sim \mathrm{B}(22, 0.6)$ and the mean, $\mathrm{E}(X) = \mu$,

a) find $\mathrm{P}(X \leq \mu)$,
b) find $\mathrm{P}(X > \mu)$.

Question 25. Given $X \sim \mathrm{B}(n, p)$ and $\mathrm{P}(X = 10) = \binom{45}{10} \times 0.25^{10} \times 0.75^{35}$ write down the mean, $\mathrm{E}(X)$, and variance, $\mathrm{Var}(X)$.

Question 26. Given $X \sim \mathrm{B}(450, p)$ and the mean, $\mathrm{E}(X)$, is 100, find the value of p

Question 27. Given $X \sim \mathrm{B}(1560, p)$ and the mean, $\mathrm{E}(X)$, is 120, find the value of p

Question 28. Given $X \sim \mathrm{B}(n, 0.3)$ and the variance, $\mathrm{Var}(X)$, is 42, find the value of n

Question 29. Given $X \sim \mathrm{B}(n, \frac{1}{6})$ and the variance, $\mathrm{Var}(X)$, is 375, find the value of n

Question 30. Given $X \sim \mathrm{B}(n, \frac{1}{8})$ and the standard deviation is 35, find the value of n

Question 31. Given $X \sim \mathrm{B}(n, \frac{5}{8})$ and the standard deviation is 105, find the value of n

Question 32. Given $X \sim \mathrm{B}(n, p)$, with $\mathrm{E}(X) = 25.2$ and $\mathrm{Var}(X) = 2.52$, find the values of n and p

Question 33. Given $X \sim \mathrm{B}(n, p)$, with $\mathrm{E}(X) = 14.4$ and $\mathrm{Var}(X) = 8.64$, find the values of n and p

Question 34. Given $X \sim \mathrm{B}(n, p)$, with $\mathrm{E}(X) = 5$ and $\mathrm{Var}(X) = \frac{60}{13}$, find the values of n and p

Question 35. Given $X \sim \mathrm{B}(100, 0.2)$ and $Y \sim \mathrm{B}(20, k)$, and that $\mathrm{Var}(X) = \mathrm{E}(Y)$, find the value of k

Question 36. It is given that $X \sim \mathrm{B}(n, p)$ and $Y \sim \mathrm{B}(k, p)$, where $0 < p < 1$. It is also given that $\mathrm{E}(X) = 3\mathrm{E}(Y)$ and $\mathrm{Var}(X) = 2\mathrm{E}(Y)$. Find the value of p

Question 37. It is given that $X \sim \mathrm{B}(n, p)$ and $Y \sim \mathrm{B}(k, p)$, where $0 < p < 1$. It is also given that $\mathrm{E}(X) = 5\,\mathrm{Var}(Y)$ and $\mathrm{E}(Y) = 3\mathrm{E}(X)$. Find the value of p

Question 38. State the conditions under which a situation can be modelled by a binomial distribution.

Question 39. A fair six-sided die is rolled until a 6 appears. Explain why we cannot use a binomial distribution to model this.

Question 40. In a small village, the probability of it raining on any one day of the year is known to be 0.22. Nico has sixteen days between breaking up and going back to school for his Easter holidays and wants to determine the probability that for at most 3 days it will be raining. Explain why it would be inappropriate to use a binomial distribution in this situation.

Question 41. Georgia attempts a crossword puzzle on the puzzle page of a newspaper every day. Over several months, Georgia has found that she is able to successfully complete the crossword puzzle 82% of the times.

a) Using the binomial distribution, find the probability that, out of a random sample of 21 days, Georgia successfully completes at least 18 crossword puzzles.

b) Using the same binomial distribution in part a), find the probability that Georgia successfully completes less than 16 crossword puzzles.

Question 42. A fair 8-sided spinner has sides numbered 1, 2, 3, 4, 5, 6, 7 and 8. The spinner is spun 35 times. The random variable X represents the number of times the spinner lands on 8.

a) Find the variance, $\text{Var}(X)$, of the number of times the spinner lands on 8

b) Find the probability that the spinner lands on an 8 at most 6 times.

c) Find $\text{P}(4 < X < 8)$.

Question 43. Emma plays snooker and is practicing a trick-shot. Each evening, she attempts the trick-shot 50 times. Every time she attempts the trick-shot, there is a probability of 0.1 that she miss-cues. The situation is modelled as a binomial distribution.

a) Find the mean number of times that Emma miss-cues in one evening.
b) Find the variance of the number of times that Emma miss-cues in one evening.
c) Find the probability that, on a particular evening, she miss-cues exactly 5 times.
d) Find the probability that, on a particular evening, she miss-cues at most 6 times.
e) Emma practices the trick-shot 50 times each evening for one week. Find the probability that she miss-cues at most 6 times on each of the 7 days.

Question 44. Curtis plays darts. He has determined the probability of him hitting the bullseye with one dart is 0.15. Curtis throws twelve darts. The situation is modelled as a binomial distribution.

a) Find the probability that Curtis will hit the bullseye more than twice.
b) Curtis throws twelve darts at three different randomly selected times. Find the probability that Curtis will hit the bullseye more than twice on one of these three times.
c) State two assumptions, in context, that would be necessary for the binomial distribution to be valid in this case.

Question 45. In a survey of 3500 A-Level Maths students across several sixth form colleges, 2240 use graphical calculators.

a) Using the binomial distribution, find the probability that, out of a random sample of 24 students, exactly 15 students have a graphical calculator.

b) Explain why the answer to part a) may not be valid for 24 students from a single class.

Question 46. A shop sells interlocking floor tiles. Tara visits the shop to purchase thirty of these tiles. It is given that 2.5% of all interlocking floor tiles of these style are faulty. By modelling the situation as a binomial distribution,

a) Find the probability that none of the tiles Tara buys are faulty.

b) Find the probability that at least three of the tiles Tara buys are faulty.

c) State two assumptions, in context, that would be necessary for the binomial distribution to be valid in this case.

Question 47. At a small sixth form college, 12% of the students belong to the cinema club. A random sample of 40 students are selected from the college. By modelling the situation as a binomial distribution,

a) Find the probability that exactly 4 students belong to the cinema club.
b) Find the probability that at least 3 but at most 8 belong to the cinema club.
c) Of the students who belong to the cinema club, 30% also belong to the book club. Assuming that these events are independent, find the probability that a randomly chosen student belongs to both the cinema club and the book club.
d) In the random sample of 40 students, find the probability that exactly one student belongs to both the cinema club and the book club.

Question 48. A type of lightbulb has a $\frac{3}{16}$ chance of being faulty. In a random sample of 15 lightbulbs, show that the probability of 2 lightbulbs being faulty is the same as the probability of 3 lightbulbs being faulty.

Question 49. Every day, Kai takes a walk in a park near his house. The probability that he spots a squirrel in the park on any particular day is 0.13

a) Find the probability that in a week Kai spots a squirrel on exactly 3 days.
b) Find the probability that Kai spots a squirrel on exactly 3 days in a week, during at most 1 of 7 randomly chosen weeks.

Question 50. Prerequisite knowledge for Q50: Conditional Probability A fair die has eight faces numbered 1 to 8

The die is rolled 40 times and the number of times an 8 is rolled is recorded. Find the probability that at least eight 8s are rolled,

a) given that fewer than eleven 8s are rolled.
b) given that more than five 8s are rolled.
c) given that more than five but fewer than eleven 8s are rolled.

Question 51. On each of the twelve floors of a large office building, there is a printer. Four of these are designated to only print colour, and the other eight only print black and white. The colour printers are in use 15% of the time and the black and white printers are in use 35% of the time. Find the probability that at any one time,

a) one of the colour printers is in use;
b) at most two black and white printers are in use;
c) none of the printers are in use;
d) exactly two of the printers are in use.

Question 52. Given $X \sim \mathrm{B}(150, 0.25)$, find the largest possible value of k such that $\mathrm{P}(X \leq k) < 0.1$

Question 53. Given $X \sim \mathrm{B}(500, 0.15)$, find the largest possible value of k such that $\mathrm{P}(X < k) < 0.1$

Question 54. In a town, it is known that 10% of local households use online food shopping. A random sample of 300 households are selected. The random variable X is defined as the number of households in the sample who use online food shopping. Calculate the largest possible value of k such that $\mathrm{P}(X < k) < 0.01$

Question 55. In a constituency, it is known that 35% of constituents traditionally vote for the independent candidate. A random sample of 400 adults in the constituency are selected. The random variable X is defined as the number of adults in the sample who voted for the independent candidate. Calculate the least possible value of k such that $\mathrm{P}(X > k) < 0.05$

Question 56. Prerequisite knowledge for Q56–Q59: Logarithms It is given that $X \sim \mathrm{B}(n, 0.07)$. Estimate the value of n given that $\mathrm{P}(X = 0) = 0.00465$

Question 57. It is given that $X \sim \mathrm{B}(n, 0.03)$. Estimate the value of n given that $\mathrm{P}(X = 0) = 0.02921$

Question 58. It is given that $X \sim \mathrm{B}(n, 0.995)$. Estimate the value of n given that $\mathrm{P}(X = n) = 0.03983$

Question 59. It is given that $X \sim \mathrm{B}(n, 0.974)$. Estimate the value of n given that $\mathrm{P}(X = n) = 0.01557$ **Use your calculator's SOLVE function for Q60 and Q61**

Question 60. It is given that $X \sim \mathrm{B}(n, 0.023)$. Estimate the value of n given that $\mathrm{P}(X \leq 1) = 0.39021$

Question 61. It is given that $X \sim \mathrm{B}(n, 0.998)$. Estimate the value of n given that $\mathrm{P}(X \geq n-1) = 0.72913$

Question 62. Prerequisite knowledge for Q62–Q63: Logarithms In a town, it is known that one person in every 15 people, on average, has the blood type B positive. A local hospital is looking for potential donors of this blood type. Determine the least number of blood donors that must be selected, at random, in order for the probability of including at least one donor that is type B positive is 0.95 or more.

Question 63. A factory has been made aware that there is a fault that affects one in every 70 components that it manufactures. To investigate, an employee is tasked with taking a random sample of components from the production line and finding a faulty component.
Determine the least number of components that must be randomly sampled in order for the probability of including at least one faulty component is 0.99 or more.